ST. XAVIER'S COLLEGE (AUTONOMOUS), KOLKATA
Department of Information Technology
B. Com Semester VI
Business Data Analytics Practical under NEP

Unit 1B & Unit 4: Introduction to Python for Business Data Analytics & Key Types of Business Data Analytics: Descriptive, Predictive and Prescriptive

Assignment 7: A Comprehensive Statistical Analysis Using Python: Data Analysis and Descriptive Analytics

Topics Covered: Descriptive Statistics, Interquartile Range (IQR); Correlation; Heatmap plot, Ordinary Least Squares Linear Regression Analysis; Hypothesis Testing

This assignment aims to familiarize students with hands-on exercises, including data analysis and insightful visualization creation. Students can understand the basics of descriptive statistics and its importance in data analysis; implementation of descriptive statistical measures using Python, emphasizing mean, median, mode, standard deviation; exploration of the concept of correlation between variables and interpret significance; regression analysis as a tool for predicting outcomes.

Descriptive Statistics

Summary Statistics: Calculate and display descriptive statistics (mean, median, mode, variance, standard deviation, etc.) for key variables in the dataset.

Data Distribution: Visualize the distribution of numerical variables using histograms or boxplots.

Correlation Analysis

Correlation Matrix: Compute the correlation matrix to identify relationships between variables.

Correlation Visualization: Create a heatmap to visualize correlations among key variables.

Regression Modelling

Simple Linear Regression: Perform a simple linear regression to predict a dependent variable using an independent variable.

Hypothesis testing is a statistical method used to make decisions about a population based on sample data. It involves formulating a null hypothesis (H_0) and an alternative hypothesis (H_1) and using test results to accept or reject H_0.

Exploratory Data Analysis involves understanding data quality, distributions, relationships, patterns, and anomalies using statistical summaries and visualizations before applying predictive models.

Students are required to create a folder as FourDigitRollno_Roomno_DeptCode_BDA (For Example, 0012_14_COMA_BDA) on their desktop. All the files related to the assignment should be saved within this folder. Students are required to use either Spyder or Jupyter Notebook to complete the assignment. The final working code should be saved as AssignmentNo_Rollno.py

Customer Data Analysis

1. A retail organization wants to analyze customer demographic and spending data to understand **income distribution**, **spending behavior**, and **relationships among key variables**. The insights will support **targeted marketing strategies and revenue optimization.** A dataset named **Customers.csv** contains customer attributes such as age, income, and spending score.

Data Import

a. Import the required libraries: **Pandas, Matplotlib, Seaborn, StatsModels**.

b. Load the dataset **Customers.csv** using the read_csv() function.

Descriptive Statistics

c. Display **basic statistical measures** (count, mean, std, min, max, quartiles) for **numerical variables.**

d. Display **summary statistics** for **categorical (non-numeric) variables.**

e. Display **descriptive statistics for the entire dataset.**

f. Drop the **non-numeric column Gender** and display the **updated information**.

g. Compute and display the following statistical measures for **numerical variables**.

- Mean
- Median
- Mode
- Standard Deviation
- Variance
- Skewness
- Kurtosis

Measures of Dispersion and Interquartile Range (IQR)

h. Calculate and display the range of the Age variable.

i. Describe the spread of Annual_Income using the five-number summary:

- Minimum (0^{th} percentile)
- First Quartile (25^{th} percentile)
- Median (50^{th} percentile)
- Third Quartile (75^{th} percentile)
- Maximum (100^{th} percentile)

j. Compute and display the Interquartile Range (IQR) of Annual_Income.

Correlation Analysis and Heatmap

k. Compute the correlation matrix for numerical variables.

l. Visualize the correlation matrix using a heatmap (seaborn.heatmap) with appropriate annotations.

Linear Regression Analysis (OLS)

m. Rename the column Annual_Income as Income and display the updated Data.

n. Perform Ordinary Least Squares (OLS) linear regression using:

- Independent Variable: Income
- Dependent Variable: Spending_Score

o. Display the **regression summary** using statsmodels.formula.api.ols.

p. Create a **scatter plot with regression line** using Seaborn's regplot() function and save the figure as **reg.png**.

Data Export

q. Save the final dataset as **Cust1.csv**.

Hypothesis Testing for Digital Marketing Campaign Performance

2. A digital marketing firm, AdWave Analytics, has launched two online advertising campaigns (Campaign A and Campaign B) for a leading fashion brand. The brand wants data-driven insights to evaluate campaign effectiveness, understand customer engagement across devices, and analyze user behavior across different age groups.
   Create a dataset as ad_campaign.csv according to the following format.

Campaign	CTR (%)	Device Type	Time Spent (sec)	Age Group
A	2.1	Mobile	120	18–25
A	2.5	Desktop	150	26–40
A	1.8	Tablet	90	41-55

A	2.9	Mobile	180	18–25
A	2.3	Desktop	160	26–40
A	2.0	Mobile	110	41-55
B	3.2	Mobile	210	18–25
B	3.6	Desktop	240	26–40
B	3.1	Tablet	200	41-55
B	3.8	Mobile	260	18–25
B	3.4	Desktop	230	26–40
B	3.0	Tablet	190	41-55
A	2.6	Tablet	140	18–25
A	2.4	Desktop	155	26–40
B	3.7	Mobile	270	18–25
B	3.5	Desktop	245	26–40
B	3.3	Tablet	205	41-55
A	2.2	Mobile	130	41-55

a. Load the CSV file.

Hypothesis Testing and Statistical Analysis

b. **Independent Samples t-Test (Campaign Performance):** Determine whether Campaign B performs better than Campaign A in terms of CTR.
   H_0: Mean CTR of Campaign A = Mean CTR of Campaign B
   H_1: Mean CTR of Campaign B > Mean CTR of Campaign A

c. **Chi-Square Test of Independence (Device & Engagement):** Examine whether customer engagement depends on device type**.** Convert CTR into engagement levels:
   High Engagement: CTR ≥ 3%
   Low Engagement: CTR < 3%
   H_0: Device type and engagement level are independent
   H_1: Device type and engagement level are dependent

d. One-Way ANOVA (Age Group & Time Spent): Test whether average time spent on the website differs across age groups.
   H_0: Mean time spent is equal across all age groups
   H_1: At least one age group has a significantly different mean time spent

Apply t-test, chi-square test, and ANOVA appropriately. Interpret statistical results.

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Exercise:

Exploratory Data Analysis (EDA)

A retail chain, RetailSphere Pvt. Ltd., wants to analyze its recent transaction data to understand customer demographics, product performance, and sales patterns. Create a dataset as retails_sales.csv according to the following structure.

Invoice_ID	**Date**	**Cust_Age**	**Gender**	**Product_Category**	**Total_Sales**	**Payment_Mode**
INV001	2024-01-05	25	Female	Clothing	2400	Card
INV002	2024-01-08	42	Male	Electronics	35000	UPI
INV003	2024-01-10	31	Female	Groceries	2000	Cash
INV004	2024-01-15	28	Male	Accessories	2400	Card

INV005	2024-01-18	55	Female	Electronics	42000	Card
INV006	2024-02-02	37	Male	Clothing	6000	UPI
INV007	2024-02-05	23	Female	Accessories	1800	Cash
INV008	2024-02-11	46	Male	Groceries	2100	UPI
INV009	2024-02-15	34	Female	Clothing	3900	Card
INV010	2024-02-20	29	Male	Electronics	38000	UPI
INV011	2024-03-03	41	Female	Groceries	1800	Cash
INV012	2024-03-10	27	Male	Accessories	3500	Card

Data Loading and Inspection

a. Import required libraries: pandas, numpy, matplotlib, seaborn.
b. Load the dataset using read_csv().
c. Display the first 5 and last 5 records.
d. Print the dataset information and shape.

Data Cleaning

e. Check for missing values and duplicates.
f. Handle missing values appropriately (if any).

Descriptive Statistics

g. Display summary statistics for numerical columns.
h. Display value counts for categorical variables.

Exploratory Data Analysis

i. Plot a histogram for Customer_Age.
j. Create a bar chart showing sales by Product_Category.
k. Draw a box plot for Total_Sales to identify outliers.
l. Plot a scatter plot between Unit_Price and Total_Sales.

Correlation Analysis

m. Compute the correlation matrix for numerical variables.
n. Visualize it using a heatmap.

Business Interpretation

o. Identify:
   - The top-selling product category
   - The most preferred payment mode
   - One key insight related to customer age or spending behavior

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